Daniel Marks

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PROFESSIONAL SUMMARY

I am a second-year DPhil candidate in Statistical Machine Learning at the University of Oxford, and a Lecturer in Computer Science at Christ Church. My research lies at the intersection of Statistical Learning Theory and Bayesian Statistics, and focuses on equipping data-driven learning algorithms and inference procedures with formal performance and safety guarantees. My work has been presented at leading academic conferences and includes an oral presentation and a best student paper award. I previously worked as an Equities Quantitative Researcher at two multi-manager hedge funds, collaborating with portfolio managers and researchers to deliver superior risk-adjusted returns.

EDUCATION

University of Oxford Oct. 2025 –
DPhil, Statistical Machine Learning

- **Thesis**: Statistical Foundations of Self-Certified Learning and Inference.
- **Supervisor**: Dr. Mark van der Wilk.

Imperial College London Oct. 2020 – Jul. 2024
MEng, Computing (AI/ML) - 1st class Hons (77%)

- **Thesis**: Naturally Self-Certified Gaussian Process Approximations via Generalised Bayesian Pick-to-Learn. **Grade: 95%, Distinguished Project Prize**. Supervisor: Dr. Dario Paccagnan, Second Marker: Dr. Yingzhen Li.
- **Relevant Modules:** Mathematics for Machine Learning (85%), Deep Learning (81%), Deep Graph-Based Learning (83%), Logic-Based Learning (94%), Computer Vision (78%), Natural Language Processing (75%).

CGS College Sep. 2018 – Jul. 2020
International Baccalaureate (IB) Diploma

- **Class Valedictorian** with a score of **45/45**.
- **Subjects**: Mathematics (HL), Computer Science (HL), Physics (HL), English (HL), History (SL), Greek (SL).

PRIZES & AWARDS

- **AISTATS Best Student Paper Award**: Awarded to the best paper with a student first author. 2025
- **Splunk Prize**: Best Master's thesis in the field of Machine Learning at Imperial College London. 2024
- **Distinguished Project Prize**: Outstanding Master's thesis in terms of technical achievement and presentation. 2024
- **Dean's List**: In the top 10% of students in the Master's year with an 85% average, at Imperial College London. 2024

FUNDING

- **G-Research Graduate Scholarship**: Fully funded DPhil scholarship in Machine Learning (competitively won). 2025
- **Imperial College Research Bursary**: Undergraduate Research Opportunity Funding (Merit-based): £4,000. 2024
- **Academic Excellence Scholarship**: Outstanding Academic Performance at CGS IB College: £20,000. 2018

PUBLICATIONS

- **Marks D**, Paccagnan D, van der Wilk M. Symmetry Guarantees Statistic Recovery in Variational Inference. The 40th Annual Conference on Neural Information Processing Systems (NeurIPS); 2026.
- Paccagnan D, **Marks D**, Campi M, Garatti S. Pick-to-Learn for Systems and Control: Data-driven Synthesis with State-of-the-art Safety Guarantees, 2025. To appear in IEEE Control Systems, vol. 46, no. 6, 2026.
- Paccagnan D, **Marks D**, Campi M, Garatti S. Pick-To-Learn for Systems and Control: theoretical review with a showcase in reachability analysis. The 64th IEEE Conference on Decision and Control (CDC); 2025.
- **Marks D**, Paccagnan D. Pick-to-Learn and Self-Certified Gaussian Process Approximations. The 28th International Conference on Artificial Intelligence and Statistics (AISTATS); 2025. **Oral** (top 2%). **Best Student Paper Award**.

RESEARCH EXPERIENCE

Research Assistant Jul. 2024 – Nov. 2024
Imperial College London - Supervisor: Dr. Matthew Wicker

- **Project**: Bayesian Mixture-of-Experts (MoE) transformer layers in LLMs.
- **Summary**: Researched Bayesian MoE layers in transformers and investigated the suitability of different approximate inference schemes (VI, HMC, MCD, KFLA) for scalable uncertainty quantification in natural language generation.

Machine Learning Research Intern Jul. 2019 – Sep. 2019
Institute for Language & Speech Processing, "Athena" Research & Innovation Centre

- **LSTM-based generative modelling** for real-time jazz improvisation accompaniment.
- **Joined a team of researchers** investigating the adaptability of RNNs for modelling complex musical structures.

PROFESSIONAL EXPERIENCE

Quantitative Analyst - Equities Dec. 2024 – Sep. 2025
North Rock Capital Management, LLC

- **Research** on crowding, pure and liquid factors, factor seasonality, thematics, net/gross flows, sector opportunity sets.
- **Earnings alpha** using alternative signals for an EU/US consumer discretionary pod.
- **Factor-neutral portfolio construction**, position sizing, hedging, and internal alpha capture.

Quantitative Research Intern Apr. 2023 – Oct. 2023
Carlson Capital L.P.

- **Trade signal research, implementation, and back-testing** for Long/Short and Relative Value equity strategies.
- **Alpha research around earnings**, revisions, and sentiment using alternative unstructured datasets.
- **Factor exposure, PnL attribution, sizing analysis** to align research with fund objectives and risk parameters.
- **Collaboration with senior PMs and Quants** to translate findings into actionable investment decisions.

TEACHING

Lecturer in Computer Science Oct. 2025 –
Christ Church, University of Oxford

- **Tutoring Prelims and Part A students** in Functional Programming, Models of Computation, Algorithms, Compilers.
- **Conducting admissions interviews** with candidates applying to read Computer Science at Christ Church.
- **Setting and marking Collections**, organising revision sessions, and providing academic and pastoral support.

Teaching Assistant Oct. 2023 – Jun. 2024
Department of Computing, Imperial College London

- **Duties**: Personal Maths Tutor (PMT) supporting a group of 10 first-year students by running weekly supervisions, marking scripts, and organising revision sessions. Reporting to: Prof. Sophia Drossopoulou.
- **Modules Supported**: Discrete Mathematics, Mathematical Logic, Graphs and Algorithms, Software Verification.

TALKS & PRESENTATIONS

- **Pick-to-Learn: Meta-learning State-of-the-art Generalisation Guarantees** Jun. 2026
  Oxford Computer Science Conference 2026, Oxford, United Kingdom. **Best Oral Presentation Award.**
- **Can We Trust Our Machine Learning Models? Avenues to Self-Certification** Oct. 2025
  Michaelmas Term Academic Soiree, Hertford College, Oxford, United Kingdom.
- **Pick-to-Learn and Self-Certified Gaussian Process Approximations** May 2025
  The 28th International Conference on Artificial Intelligence and Statistics (AISTATS), Mai Khao, Thailand.
- **Tight Risk Certificates for Black Box Learners through Preferent Compression** Apr. 2025
  London Operations Research Day (LORD), Worcester College, Oxford, United Kingdom.

OTHER SKILLS

- **Classical Piano Diploma (LRSM equiv.)**: 100/100, First Prize, félicitations du jury.
- **Languages**: English (Native), Greek (Native), French (DELF B2).